

Project Title: CNG 476 System Simulation Final Report

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GitHub: <https://github.com/muratdgunes/System-Simulation-Project/>

Abstract—This project presents a discrete-event simulation study of a LoRa-based Smart Home Monitoring and Emergency Alert System. The simulated system consists of six IoT sensor nodes, one LoRa gateway, a central controller, a shared LoRa wireless channel, and a gateway-side priority queueing mechanism. Temperature and humidity sensors generate routine low-priority monitoring traffic, while smoke alarm sensors and door sensors generate high-priority emergency and security traffic. The simulation was implemented using OMNeT++, INET, and FLoRa. The main objective was to evaluate packet delivery reliability, end-to-end delay, received packet volume, and gateway queue behaviour under normal and emergency-heavy traffic conditions. Two scenarios were analysed: a Base Scenario representing normal smart home operation and an Emergency Constant Burst Scenario representing sustained high traffic. In the Base Scenario, the system achieved 100.0% Packet Delivery Ratio with a mean end-to-end delay of 0.1694 s and a 95% confidence interval of ± 0.0018 s. In the Emergency Constant Burst Scenario, the Packet Delivery Ratio decreased to 60.3%, while the mean delay increased to 2.6665 s with a 95% confidence interval of ± 0.1334 s. The results show that the proposed LoRa-based system performs reliably under normal traffic conditions, but sustained burst traffic increases packet loss, delay, and gateway queueing pressure.

Index Terms—System simulation, OMNeT++, INET, FLoRa, IoT, discrete-event simulation.

I. INTRODUCTION

Smart home monitoring systems are increasingly important for improving resident safety and awareness of emergency. Recent IoT based home safety systems also use different sensors to detect risks such as fires, gas leaks and abnormal environmental conditions in real time [8]. Systems like this, different Internet of Things (IoT) devices continuously observe the environment of houses and report useful information to a central controller. Routine sensors like temperature and humidity sensors, generate periodic monitoring data while emergency based devices such as smoke alarm sensors and door sensors generate event focused alarm traffic. Since emergency messages may represent critical emergency situations or security breaches like unauthorized door activity, they must be delivered with low delay and high reliability.

This project focuses on the simulation of a Smart Home Monitoring and Emergency Alert System using a LoRa-based IoT network. The simulated system consists of six IoT sensor nodes, one LoRa gateway, a network server/central controller,

a LoRa wireless channel, and a gateway-side priority queueing mechanism. Temperature and humidity sensors represent routine low-priority monitoring traffic, whereas smoke alarm sensors and door sensors represent high-priority emergency or security-related traffic. The main challenge is that both routine and emergency packets share the same communication infrastructure and are forwarded through the same gateway. Therefore, packet accumulation at the gateway and possible delay in emergency packet delivery must be examined carefully.

Simulation is necessary because the behaviour of this system depends on random event occurrences, wireless communication effects, packet forwarding delays, and queueing behaviour at the gateway. In a real smart home environment, emergency events do not occur at fixed deterministic times. Door events and smoke alarm events are better represented as stochastic processes, while routine monitoring packets are generated periodically. A discrete-event simulation approach is suitable for this problem because the system state changes mainly when events occur, such as packet generation, wireless transmission, packet reception, queue insertion, or packet forwarding.

The purpose and main objective of this project is to model the LoRa-based monitoring systems that used in smart homes, implement the sensor traffic generation and gateway sided queueing mechanism and evaluate the system performance under the base scenario. The main performance metrics are Packet Delivery Ratio, End-to-End Delay, Gateway Queue Length, and Total Received Packets. We use these metrics to understand if the network can send monitoring and emergency packets successfully. We also use them to see if the gateway can handle the traffic without a long waiting problem.

II. SELECTED PROJECT DESCRIPTION

The selected project is titled *Smart Home Monitoring and Emergency Alert System*. This project models a LoRa based IoT smart home system. In this system, different sensors monitor the home and send their data to a central controller through a LoRa gateway. The main real life purpose of this system is to support reliable home monitoring and fast emergency notification. Normal environmental data and emergency alarm messages use the same communication structure. Because of

TABLE I
SELECTED PROJECT SUMMARY

Item	Description
Selected Project Title	Smart Home Monitoring and Emergency Alert System
Application Domain	IoT based smart home monitoring and emergency alerting
Main Problem	Routine monitoring traffic and high priority emergency traffic use the same LoRa based communication structure. This may cause gateway queueing and packet delay.
Simulation Objective	To model the LoRa based smart home monitoring system and evaluate reliability, delay, packet reception and gateway queue behaviour under the base traffic configuration.
Main Users/Entities	Temperature and humidity sensors, smoke alarm sensors, door sensors, LoRa gateway, network server or central controller and smart home monitoring users.
Main Performance Metrics	Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets.

this, the system should be checked in terms of reliability, delay and gateway queue behaviour.

The simulated system has six IoT sensor nodes. It also has one LoRa gateway, one network server or central controller, one LoRa wireless channel and a priority queueing mechanism at the gateway side. Nodes 0 and 1 represent temperature and humidity sensors. These nodes generate routine monitoring traffic with low priority. Nodes 2 and 3 represent smoke alarm sensors. They generate emergency alarm traffic with high priority. Nodes 4 and 5 represent door sensors. They generate security event traffic with high priority. The gateway separates high priority packets and low priority packets into two logical queues. Both queues are served by the same gateway processor.

The main problem in this project is that routine monitoring packets and emergency packets use the same LoRa based smart home network. If many sensors generate packets close to each other, packets may wait at the gateway. This can increase end to end delay. This problem is more important for smoke alarm sensors and door sensors. If emergency messages are delayed or lost, the safety value of the monitoring system can decrease. For this reason, the project focuses on sensor traffic modelling, priority queueing at the gateway and communication performance under the base smart home traffic configuration.

The purpose of this simulation study is to check whether the LoRa based smart home monitoring system can deliver sensor packets reliably. It also checks whether the gateway can process the generated traffic without a long congestion problem. The simulation is implemented as a discrete event simulation by using OMNeT++, INET and FLoRa. The main performance metrics are Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets.

III. SYSTEM ARCHITECTURE

The system architecture of the proposed Smart Home Monitoring and Emergency Alert System is shown in Fig. 1. This architecture represents a LoRa based IoT smart home system. In this system, different sensors communicate with a central controller through a LoRa Home Gateway. The network uses

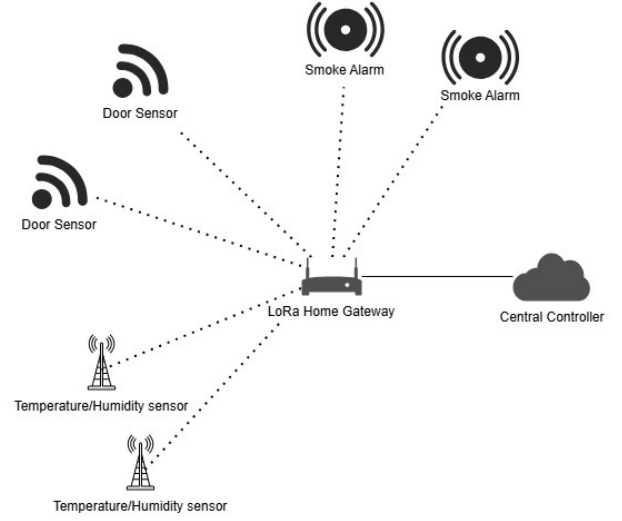


Fig. 1. System Architecture of the Smart Home Monitoring and Emergency Alert System

a star topology structure. This means that all sensor nodes send their packets to one gateway by using the shared LoRa wireless medium.

The implemented system has six IoT sensor nodes. These nodes are divided into three groups according to their tasks. Nodes 0 and 1 represent temperature and humidity sensors. They generate routine monitoring traffic with low priority at regular times. Nodes 2 and 3 represent smoke alarm sensors. They generate emergency alarm traffic with high priority. Nodes 4 and 5 represent door sensors. They generate security related traffic with high priority. These sensor nodes communicate with the LoRa Home Gateway by using LoRa communication provided through the FLoRa framework.

The LoRa Home Gateway is the main communication and forwarding point of the system. The gateway receives packets from all sensor nodes and forwards them to the Central Controller. The gateway also separates packets according to their priority. For this purpose, two queue structures are used. One queue is used for high priority packets and the other one is used for low priority packets. Emergency packets from smoke alarm sensors and door sensors are handled in the high priority queue. Routine monitoring packets from temperature and humidity sensors are handled in the low priority queue. Both queues share the same gateway side processing server.

The Central Controller represents the network server and monitoring part of the smart home system. It collects sensor data and receives emergency alarms. It also records the simulation outputs used in the performance analysis. This architecture is designed to evaluate mixed priority IoT traffic in a LoRa based wireless network. It also helps to observe gateway side queueing conditions.

IV. SYSTEM MODEL AND ASSUMPTIONS

The simulated system is modelled as a LoRa based discrete event IoT network. It is used for smart home monitoring and emergency alerting. The system has six IoT sensor nodes, one LoRa Home Gateway and one network server or Central Controller. It also uses a shared LoRa wireless communication channel. The sensor nodes create different packet types according to their roles in the smart home. Temperature and humidity sensors create routine monitoring packets. Smoke alarm sensors and door sensors create emergency or security related packets.

The communication flow starts from the sensor nodes. When a sensor creates a packet, the packet is sent to the LoRa Home Gateway through the LoRa wireless channel. After the packet reaches the gateway, it is placed into a logical queue according to its priority level. Emergency packets from smoke alarm sensors and door sensors are placed into the high priority queue. Routine packets from temperature and humidity sensors are placed into the low priority queue. The gateway always serves high priority packets first. After that, it serves low priority packets. However, both queues use the same gateway side processing server. Because of this, the gateway queueing structure is shown as an M/M/1 style model with priority separation. This priority based queueing idea is similar to M/M/1 queueing models with two priority classes, where one class has priority over the other class [13].

The Central Controller represents the network server and the final monitoring point of the system. It receives the packets forwarded by the gateway. It also records the performance results used in the analysis. The main state variables of the simulation are packet generation status, packet priority, queue length, packet reception count, packet delay and packet delivery status. The main events are packet generation, wireless transmission, packet arrival at the gateway, queue insertion, gateway service, packet forwarding and packet reception at the Central Controller.

The traffic generation model changes according to the sensor type. Temperature and humidity sensors generate deterministic periodic packets with a fixed one hour interval. Their first packet generation time is selected randomly between 0 and 1 hour by using a uniform distribution. This prevents all routine sensors from sending their first packet at the same time. According to Renkeer (2026), using a sensor which is updated once an hour stay active for 1 year or more. On the other hand, sensors updated in short intervals such as five minutes, stay active for approximately 6 months. This is the main reason why we selected our temperature/humidity sensors to send packets every 1 hour. With this configuration, we tried to make our simulation as realistic as possible. Smoke alarm sensors generate high priority emergency packets with exponentially distributed inter arrival times. The mean value is 24 hours. We selected interval value is based on National Fire Protection Association (2013), supervisory signal reactivation practices. These signals are activated every 24 hours, in context of fire alarms, this source helped us to determine our fire alarm mean

TABLE II
SIMULATION ENTITIES, EVENTS, AND STATE VARIABLES

Entity	Events	State Variables
Temperature/Humidity Sensors	Periodic packet generation and LoRa transmission	Packet generation time, packet type, packet priority and sent packet count
Smoke Alarm Sensors	Emergency packet generation and LoRa transmission	Inter arrival time, packet priority, sent packet count and emergency packet status
Door Sensors	Security event packet generation and LoRa transmission	Inter arrival time, packet priority, sent packet count and event packet status
LoRa Wireless Channel	Wireless packet transmission and reception attempt	Transmission state, reception state and channel effects
LoRa Home Gateway	Packet arrival, queue insertion, service start, service completion and packet forwarding	High priority queue length, low priority queue length, server state and packet waiting status
Central Controller / Network Server	Packet reception and output recording	Received packet count, packet delay, packet delivery status and recorded simulation metrics

inter arrival time. Door sensors generate high priority security event packets with exponentially distributed inter arrival times. For door sensors, the mean value is 4 hours. Which is retrieved from Washington State University article "Using continuous sensor data to formalize a model of in-home activity patterns". In this source, figure 11 states that, Personal Hygiene inter-arrival times range from 0-6 and 3-18 hours. This is why we configured our door sensors, for example in bathroom, as exponential distribution with mean of 4 hours. Additionally, our simulation time is selected as 72 hours which is three times greater than our slowest sensor, smoke alarm, triggering with mean inter arrival time of 24 hours. According to Omnet++ simulation methodology, simulation should go on until stable results are achieved rather than fixed time. By configuring 72 hours as our simulation time, we successfully reach stable results by surpassing 24 hour exponential sensor three times. Lastly, according to Pawlikowski(1990), to remove initialization bias of simulation data we employed warmup parameter. This parameter is configured as 2 hours, which safely helps us to reach stable state by letting our temperature/humidity sensors to send packets to the gateway two full cycles. Which helps network queues and LoRaWAN gateway to be initialized.

The main simulation entities, their events and state variables are summarized in Table II. The main modelling assumptions and their technical reasons are given in Table III.

V. SIMULATION METHODOLOGY

The proposed smart home monitoring system is modelled with a discrete event simulation method. This method is suitable for the project because the system state does not change all the time. It changes only when a specific event happens. Important events are packet generation at sensor nodes and wireless transmission over the LoRa channel. Other important events are packet arrival at the LoRa Home Gateway, queue insertion, gateway service, packet forwarding and packet reception at the Central Controller. Between these events, the simulation clock moves directly to the next scheduled event.

TABLE III
MODELLING ASSUMPTIONS AND JUSTIFICATION

Assumption	Justification
The smart home network is represented by six IoT sensor nodes, one LoRa gateway and one Central Controller.	This model includes the main communication parts needed for monitoring and emergency alert delivery. It avoids extra physical home details.
Temperature and humidity sensors generate low priority routine traffic every 1 hour.	Environmental monitoring is periodic. It is less urgent than emergency alarm traffic.
Smoke alarm sensors generate high priority traffic using exponential inter arrival times with a mean of 24 hours.	Smoke alarm events are random emergency events. Because of this, stochastic arrivals are more suitable than fixed periodic updates.
Door sensors generate high priority traffic using exponential inter arrival times with a mean of 4 hours.	Door events happen randomly over time. They may also show security related activity, so they are modelled as high priority random events.
The first packet time of routine sensors is randomized between 0 and 1 hour.	This prevents routine sensors from starting at exactly the same time. It also prevents artificial synchronization.
The LoRa gateway uses two logical queues. One queue is high priority and one queue is low priority.	Logical queue separation helps the gateway distinguish emergency traffic and routine traffic.
The gateway always serves the high priority queue before the low priority queue.	Emergency and security packets should be processed before routine monitoring packets in a safety focused smart home system.
Both logical queues share a single gateway side server.	The gateway has one processing and forwarding point. Because of this, the queueing structure follows an M/M/1 style model with priority separation.
The LoRa wireless channel is shared by all sensor nodes.	In the implemented LoRa based network, all sensor nodes communicate with the gateway through the same wireless medium.
The emergency case is modelled as a constant burst over the 3 day simulation period instead of a short sudden burst.	This creates a long high traffic emergency condition. It also helps to observe gateway behaviour during the full simulation time.
Physical indoor obstacles and detailed room geometry are not modelled directly.	The focus of the project is traffic generation, LoRa communication, packet delivery, delay and gateway queueing behaviour. It is not detailed indoor propagation modelling.

The simulation uses a LoRa based IoT network model. It is implemented in OMNeT++, INET and FLoRa. OMNeT++ provides the event driven simulation engine. INET supports the network communication parts. FLoRa provides LoRa and LoRaWAN communication modelling. The simulation configuration uses `cMersenneTwister` as the random number generator. The simulation duration is set to 3 days. A 2 hour warm up period is used before collecting the final performance results. To reduce the effect of randomness, the base scenario is executed for 20 independent runs.

The packet generation process depends on the sensor type. Temperature and humidity sensors generate routine monitoring traffic. This traffic is deterministic and periodic. Their first packet generation time is selected randomly between 0 and 1 hour by using a uniform distribution. This prevents artificial synchronization at the start of the simulation. After the first packet, these sensors generate packets every 1 hour. Smoke alarm sensors generate high priority emergency packets by using an exponential inter arrival time distribution. The mean value is 24 hours. Door sensors generate high priority security event packets by using an exponential inter arrival time distribution. The mean value is 4 hours.

The exponential distribution is used for smoke alarm and

TABLE IV
RANDOM PROCESSES AND PROBABILITY MODELS

Process	Distribution / Model	Purpose in the Simulation
Initial routine packet generation	Uniform distribution between 0 and 1 hour	Randomizes the first packet time of temperature and humidity sensors. This avoids synchronized initial transmissions.
Routine monitoring packet generation	Deterministic 1 hour interval	Models regular temperature and humidity monitoring traffic.
Smoke alarm packet generation	Exponential inter arrival time with mean 24 hours	Models random high priority emergency alarm events.
Door sensor packet generation	Exponential inter arrival time with mean 4 hours	Models random high priority security event traffic.
Gateway queue selection	Priority based logical queue selection	Gives service priority to emergency packets before routine packets.
Gateway service / forwarding	Single shared gateway server model	Represents packet forwarding from the LoRa gateway to the Central Controller. It does not add an extra service time distribution.
Monte Carlo replications	20 independent simulation runs	Reduces the effect of random variation and supports statistical output analysis.

door sensor events because these events happen randomly over time. They are not controlled by a fixed schedule. On the other hand, temperature and humidity readings are routine monitoring updates. Because of this, they are represented with deterministic periodic generation. This separation helps the simulation represent both normal background traffic and random high priority traffic in the same LoRa based smart home network.

Gateway side queueing is handled with two logical queues and one shared processing server. Packets from smoke alarm sensors and door sensors are inserted into the high priority queue. Packets from temperature and humidity sensors are inserted into the low priority queue. When the gateway is ready to forward a packet, it first checks the high priority queue. If there is no high priority packet waiting, it then serves the low priority queue. This event driven queueing mechanism helps the simulation observe the effect of mixed priority traffic on delay and gateway queue length.

The simulation is not modelled as a time stepped system. In a time stepped model, the full system state would need to be checked at every fixed time interval. This would also happen when no sensor activity occurs. This would be inefficient for this IoT application because packets are generated only at specific event times. The project is also not trace driven. This is because the input traffic is generated by probability models and configured packet generation rules. It is not taken from a fixed external traffic trace.

The random processes and probability models used in the simulation are summarized in Table IV. The main scheduled events in the discrete event simulation are summarized in Table V.

VI. OMNeT++, INET, AND FLoRA IMPLEMENTATION

The final simulation project was implemented using OMNeT++, INET, and FLoRa. OMNeT++ was used as the

TABLE V
SCHEDULED EVENTS IN THE DISCRETE EVENT SIMULATION

Scheduled Event	Trigger Condition	State Change / Action
Initial packet generation	Sensor application starts and the first packet timer expires	A sensor creates its first packet according to its traffic model.
Periodic routine packet generation	One hour timer expires for temperature and humidity sensors	A low priority routine packet is generated. Then the next packet event is scheduled.
Smoke alarm packet generation	Exponential inter arrival timer expires for a smoke alarm sensor	A high priority emergency packet is generated. Then the next smoke alarm event is scheduled.
Door sensor packet generation	Exponential inter arrival timer expires for a door sensor	A high priority security packet is generated. Then the next door sensor event is scheduled.
LoRa wireless transmission	A generated packet is ready to be transmitted	The packet is sent through the shared LoRa wireless channel toward the gateway.
Packet arrival at gateway	The gateway successfully receives a packet from the LoRa channel	The packet is classified according to its priority and inserted into the correct logical queue.
Gateway service selection	The shared gateway server becomes available	The gateway first checks the high priority queue. If it is empty, the low priority queue is served.
Packet forwarding to controller	Gateway service or forwarding action is completed	The packet is forwarded from the gateway to the Central Controller or network server.
Packet reception at controller	The Central Controller receives a forwarded packet	Received packet count, packet delay and delivery related output statistics are updated.

main discrete-event simulation environment, INET provided the supporting network communication structure, and FLoRa was used for modelling the LoRa-based wireless IoT communication between the sensor nodes and the gateway. The final configuration file used for the experiments is `n100-gw1.ini`. This configuration defines the implemented network as `flora.simulations.LoRaNetworkTest`, with six LoRa sensor nodes and one LoRa gateway.

The implementation uses FLoRa LoRa nodes to represent the smart home sensors. Nodes 0 and 1 are configured as temperature and humidity sensors that generate routine monitoring traffic. Nodes 2 and 3 are configured as smoke alarm sensors, and Nodes 4 and 5 are configured as door sensors. The application type used at the sensor nodes is `SimpleLoRaApp`. The gateway forwards packets to the network server, where the server-side application is configured as `NetworkServerApp`. Adaptive Data Rate evaluation is enabled on both the node side and server side, because ADR in LoRaWAN is used to adapt transmission parameters such as data rate and transmission power to improve network performance and transmission efficiency (Garg, 2019). The LoRa wireless channel uses the `LoRaLogNormalShadowing` path loss model.

The main custom implementation was added to the packet forwarding and network server components. The priority queueing mechanism was implemented in `PacketForwarder.h` and `PacketForwarder.cc`. The gateway maintains two logical queues: a high-priority queue and a low-priority queue. Packets generated by smoke alarm sensors and door sensors are inserted into the high-priority queue, while routine packets generated by temperature and humidity sensors

are inserted into the low-priority queue. When the gateway forwards packets, it always checks and serves the high-priority queue first. If the high-priority queue is empty, the gateway then serves the low-priority queue. Since both queues share one gateway-side forwarding server, this implementation follows the intended M/M/1-style gateway abstraction with priority separation.

The project also includes custom metric recording. The files `PacketForwarder.ned` and `NetworkServerApp.ned` were modified by adding `@signal` and `@statistics` definitions for the required output metrics. These NED-level metric declarations were connected to the C++ implementation logic. In `NetworkServerApp.cc` and `NetworkServerApp.h`, a `delaySignal` mechanism was added to record end-to-end packet delay. In `SimpleLoRaApp.cc`, the packet-related metric values were formatted so that sent-packet statistics could be recorded correctly. The final configuration records custom vectors for high-priority queue length, low-priority queue length, and delay, while scalar outputs are used for sent packets, received packets, and packet drop information.

The packet flow begins when a sensor application generates a packet according to its configured traffic model. The packet is transmitted through the LoRa wireless medium and received by the LoRa gateway if the wireless transmission is successful. Then, the gateway classifies the packet according to its priority and inserts it into the corresponding logical queue. The shared gateway server forwards packets toward the network server. Finally, the `NetworkServerApp` records received-packet information and end-to-end delay values for output analysis.

The output figures used in the report were generated using Python scripts after the simulation outputs were collected. The script `plot_graph.py` was used to generate the delay histogram. The script `plot_drop.py` was used to generate the 95% confidence interval boxplot. The script `plot_packets.py` was used to create the Packet Delivery Ratio proof figure using the processed sent and received packet totals. The script `plot_queue.py` was used to generate the gateway queue length timeline.

The simulation was executed successfully using the final configuration. Fig. 2 shows the beginning of the OMNeT++ Cmdenv execution, where the NED files are loaded, the configuration is prepared, and the `LoRaNetworkTest` scenario is initialized. Fig. 3 shows the completion of the same simulation run, where the execution reaches 100% progress and terminates after the configured simulation time limit. These screenshots provide execution evidence that the submitted simulation configuration can be launched and completed successfully.

VII. MAPPING OF COURSE TOPICS TO THE PROJECT

This section explains how the main topics in CNG 476 System Simulation are used in the final project. The project is not only a LoRa network implementation. It is also a discrete event simulation study. It includes system modelling, random

Fig. 2. Simulation Execution Start Evidence

Fig. 3. Simulation Execution Completion Evidence

VIII. EXPERIMENTAL DESIGN

Each simulation experiment was run for 3 days. A 2 hour warm up period was also used. This warm up period helps to reduce the effect of the first unstable part of the simulation before the final results are collected. The experiments were repeated for 20 independent runs by using the `cMersenneTwister` random number generator. Repeated runs were needed because door sensor and smoke alarm packet generation times are random.

File Name	File Type	Purpose
n100-gw1.ini	INI	Defines the final simulation configuration, including the network, number of nodes, number of gateways, simulation duration, warm-up period, repetitions, RNG class, traffic generation models, LoRa parameters, and recorded outputs.
PacketForwarder.ned	NED	Defines gateway-side packet forwarding structure and includes custom <code>@signal</code> and <code>@statistics</code> definitions for queue-related metrics.
NetworkServerApp.ned	NED	Defines network server application statistics and signal declarations for server-side output recording.
PacketForwarder.cc	C++	Implements the gateway-side packet forwarding logic, including high-priority and low-priority queue handling.
PacketForwarder.h	Header	Declares the data structures and functions required for the gateway-side priority queueing mechanism.
SimpleLoRaApp.cc	C++	Handles sensor-side packet generation behaviour and formats packet-related metrics for correct output recording.
NetworkServerApp.cc	C++	Implements server-side packet reception logic and records end-to-end packet delay using <code>delaySignal</code> .
NetworkServerApp.h	Header	Declares the server-side variables and signal structures required for delay and reception metric recording.
plot_graph.py	Python	Processes delay output and generates the delay histogram figure.
plot_drop.py	Python	Generates the received-packet boxplot with 95% statistical interpretation.
plot_packets.py	Python	Creates the Packet Delivery Ratio proof graph using processed sent and received packet totals.
plot_queue.py	Python	Generates the gateway queue length timeline for high-priority and low-priority queues.

Tool / Framework	How It Is Used in the Project
OMNeT++	Used as the main discrete-event simulation environment. It manages the simulation clock, event scheduling, repeated runs, configuration handling, vector/scalar recording, and execution of the implemented modules.
INET	Used as the supporting network framework for communication-related components and gateway-to-server packet forwarding structure. It provides the network simulation foundation required by the implemented OMNeT++/FLoRa project.
FLoRa	Used to model LoRa-based wireless IoT communication between the sensor nodes and the LoRa gateway. It supports LoRa node applications, gateway behaviour, LoRa wireless channel modelling, Adaptive Data Rate evaluation, and LoRa path loss modelling.

Two scenarios were defined for the experimental study. The first scenario is the *Base Scenario*. This scenario represents normal smart home operation. In this scenario, temperature and humidity sensors generate low priority routine packets every 1 hour. Their first packet time is selected randomly between 0 and 1 hour. Smoke alarm sensors generate high priority packets by using exponentially distributed inter arrival times. The mean value is 24 hours. Door sensors also generate

TABLE VIII
MAPPING OF CNG 476 COURSE TOPICS TO THE FINAL PROJECT

Course Topic	Project Specific Implementation
Simulation methodology and modelling	The smart home monitoring system is modelled as a discrete event simulation. The system has clear entities, events, states and performance goals. The main entities are six IoT sensor nodes, one LoRa Home Gateway, one Central Controller and the LoRa wireless medium. The system state changes when events happen. These events include packet generation, transmission, gateway arrival, queue insertion, service, forwarding and reception.
System entities, attributes, and state variables	Each sensor node is defined as an entity in the project. The important attributes are sensor type, traffic priority, packet generation model, spreading factor, transmission power and node position. Important state variables include high priority queue length and low priority queue length. Other state variables are server state, sent packet count, received packet count, packet delay and packet delivery status.
Discrete event simulation	The system is dynamic and event driven. This is because packets are generated and processed at specific points in simulation time. The simulation clock does not move with fixed time steps. Instead, OMNeT++ moves the simulation from one scheduled event to the next one. This matches the LoRa based IoT network because the system mainly changes when packets are generated, transmitted, queued or received.
Event scheduling	Event scheduling is used for sensor packet generation and gateway processing. Temperature and humidity sensors schedule packet generation every 1 hour after a randomized first packet time. Smoke alarm sensors schedule high priority events by using <code>exponential(24h)</code> . Door sensors schedule high priority events by using <code>exponential(4h)</code> . Gateway forwarding events start when packets are available and the shared gateway server can process them.
Random number generation	The simulation uses pseudo random number generation with <code>cMersenneTwister</code> in the OMNeT++ configuration. Random numbers are needed for the first packet time of routine sensors. They are also needed for the random inter arrival times of smoke alarm and door sensor packets. This supports independent, reproducible and controlled simulation runs.
Random variate generation	Random variate generation is used through OMNeT++ distribution functions. The project uses <code>uniform(0h,1h)</code> for randomized first packet times of periodic sensors. It also uses <code>exponential(24h)</code> and <code>exponential(4h)</code> for emergency and security events. These generated values are used as simulation inputs.
Probability models	The project uses probability models to represent uncertainty in sensor event timing. Smoke alarm and door sensor events are modelled with exponential inter arrival times. This is because they represent random events over time. Temperature and humidity traffic is deterministic and periodic because routine environmental sensing happens at regular times.
Stochastic processes	Door sensor and smoke alarm packet arrivals are represented as stochastic processes over simulation time. Their event times are random. Because of this, the number of generated events can change between simulation runs. This stochastic behaviour affects received packet counts, delay results and queue activity.
Poisson arrival process interpretation	The exponential inter arrival time assumption for smoke alarm and door sensor packets is related to the Poisson arrival model taught in the course. In this approach, independent random arrivals over time can be represented with exponentially distributed inter arrival times. In this project, this idea is used to justify random emergency and security event generation.
Queueing analysis	The LoRa Home Gateway is the main queueing point in the system. It has two logical queues. One queue is for high priority smoke alarm and door sensor packets. The other queue is for low priority temperature and humidity packets. Both queues share one gateway side server. The gateway always serves the high priority queue before the low priority queue. This allows the project to analyse priority based queueing behaviour with an M/M/1 style single server model.
Monte Carlo simulation	The final scenario is executed for 20 independent runs. Door and smoke alarm events are stochastic. Because of this, one simulation run may not show the average behaviour of the system. Repeated runs reduce single run randomness. They also support statistical output analysis such as mean delay, standard deviation and confidence interval calculation.
Input analysis	The input model is explained by using the real meaning of each traffic source. Temperature and humidity monitoring is periodic and deterministic. Door and smoke alarm events are random over time. Since no formal Q-Q plot or goodness of fit test is used, the report gives mathematical and physical justification for the selected input distributions. It does not claim formal distribution fitting.
Output analysis	The project records simulation outputs by using OMNeT++ vectors and scalars. The main output metrics are Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets. The final processed results include total sent packets, total received packets, PDR, delay mean, delay standard deviation and a 95% confidence interval for delay.
Confidence interval and statistical interpretation	The delay metric is analysed statistically by using the processed output values from 20 runs. The final delay mean is reported with standard deviation and a 95% confidence interval half width. This gives a more reliable interpretation than using only one simulation run.
Time stepped and trace driven comparison	The project uses a discrete event approach instead of a time stepped approach. This is because sensor events happen at scheduled packet generation times. They do not happen at every fixed time interval. The project is also not trace driven. This is because the input traffic is generated from probability models and simulation parameters. It is not taken from an external real world trace file.
OMNeT++ implementation	OMNeT++ is used as the main discrete event simulation engine. It manages the simulation clock, scheduled events, repeated runs, configuration parameters, vector and scalar output recording and Cmdenv execution. The implementation uses NED files, C++ files, header files, Python plotting scripts and the final <code>n100-gw1.ini</code> configuration file.
INET integration	INET is used as the supporting network simulation framework together with OMNeT++ and FLoRa. It provides reusable network structure such as modules, message passing, communication support, statistics and simulation execution structure. These parts help connect the LoRa gateway and the server side components.
FLoRa integration	FLoRa is used as the LoRa specific simulation layer. It provides LoRa nodes, gateways, LoRa medium behaviour, Adaptive Data Rate support, path loss modelling and network server behaviour. In this project, FLoRa helps to model LoRa based communication between the sensor nodes and the gateway.
Performance evaluation	The project evaluates system performance with measurable metrics. It does not only show that the simulation runs. Packet Delivery Ratio measures reliability. End to End Delay measures timeliness. Gateway Queue Length shows congestion behaviour. Total Received Packets shows the received traffic amount across simulation runs.

high priority packets by using exponentially distributed inter arrival times. For door sensors, the mean value is 4 hours. This scenario is used to observe the system under normal monitoring traffic and occasional emergency traffic.

The second scenario is the *Emergency Constant Burst Scenario*. This scenario uses the same number of nodes and gateways. However, the packet generation intervals are

much shorter. This creates high traffic during the full 3 day simulation period. In this scenario, all nodes use an initial packet time with `uniform(0s,10s)`. They also use `exponential(300s)` as the next packet interval in the general burst configuration. For emergency nodes, which are Nodes 2, 3, 4 and 5, the traffic intensity is increased more. Their first packet time is set to `uniform(0s,2s)`. Their

TABLE IX
EXPERIMENTAL DESIGN SUMMARY

Item	Value / Description
Simulation Duration	3 days
Warm-up Period	2 hours
Number of Runs	20 independent replications
Random Number Generator	cMersenneTwister
Network Configuration	6 IoT sensor nodes, 1 LoRa gateway, 1 Central Controller and shared LoRa wireless channel
Baseline Scenario	Base Scenario with routine low priority monitoring traffic and occasional high priority emergency traffic
Alternative Scenario	Emergency Constant Burst Scenario with shorter packet generation intervals for routine and emergency traffic
Base Scenario Traffic Model	Nodes 0-1: <code>uniform(0h,1h)</code> first packet and 1 hour periodic interval. Nodes 2-3: <code>exponential(24h)</code> . Nodes 4-5: <code>exponential(4h)</code>
Emergency Constant Burst Traffic Model	All nodes: <code>uniform(0s,10s)</code> first packet and <code>exponential(300s)</code> next packet interval. Nodes 2-5: <code>uniform(0s,2s)</code> first packet and <code>exponential(15s)</code> next packet interval
Varied Parameters	Packet generation interval and traffic intensity
Collected Metrics	Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets
Output Figures	Delay histogram, received packet boxplot, Packet Delivery Ratio graph and gateway queue length timeline

next packet interval is set to `exponential(15s)`. Because of this, the scenario represents a constant emergency burst condition, not only a short sudden burst.

The main changed parameter between the two scenarios is packet generation intensity. The Base Scenario uses more realistic routine and event based traffic intervals. The Emergency Constant Burst Scenario uses much shorter packet inter arrival times. This increases the load on the LoRa gateway and the wireless channel. The collected metrics are Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets. These metrics are used to evaluate reliability, delay, gateway congestion and received traffic amount.

IX. PERFORMANCE METRICS

The performance of the Smart Home Monitoring and Emergency Alert System is evaluated with four main metrics. These metrics are Packet Delivery Ratio, End to End Delay, Gateway Queue Length and Total Received Packets. They were selected because they show important parts of the LoRa based smart home network. They help us understand reliability, delay, gateway waiting behaviour and received traffic amount. Since this project is related to emergency monitoring, packet loss and packet delay are very important. Alarm messages should reach the Central Controller successfully and quickly.

The communication reliability is measured by the Packet Delivery Ratio (PDR). It is the packet delivery ratio to the network server. The first packets are generated by the sensor nodes. The higher the packet delivery ratio, the more packets were successfully delivered to the Central Controller. The PDR is calculated as follows:

$$PDR = \frac{\text{Total Received Packets}}{\text{Total Sent Packets}} \times 100 \quad (1)$$

TABLE X
PERFORMANCE METRICS

Metric	Unit	Definition / Purpose
Packet Delivery Ratio	%	Measures the percentage of generated packets that are received by the network server. It is used to evaluate communication reliability.
End to End Delay	seconds	Measures the time between packet generation at the sensor node and packet reception at the Central Controller. It is used to evaluate delay, especially for emergency packets.
Gateway Queue Length	packets	Measures the number of packets waiting in the gateway queues. High priority and low priority queue lengths are recorded separately to observe gateway congestion.
Total Received Packets	packets	Measures the total number of packets received by the network server. It is used to compare received traffic amount in different runs and scenarios.
Packet Loss Count	packets	Measures the difference between total sent packets and total received packets. It is used as supporting information for Packet Delivery Ratio.

End to End Delay measures the total time between packet generation and packet reception. The packet is first generated at a sensor node. Then it reaches the Central Controller. This metric is important because emergency packets should be delivered with low delay. Smoke alarm and door sensor messages are examples of these packets. In the implementation, delay values are recorded with the `delaySignal` output at the network server.

Gateway Queue Length measures the number of packets waiting at the LoRa Home Gateway. The gateway is the main forwarding point between the LoRa sensor nodes and the Central Controller. Because of this, queue length helps us see whether packets accumulate under different traffic conditions. The implementation records both `highQueueLength` and `lowQueueLength`. This makes it possible to analyse high priority and low priority traffic separately.

Total Received Packets measures the number of packets successfully received by the network server during the simulation. This metric is used to compare the received traffic amount in different runs and scenarios. It is also used together with the total sent packet count to calculate Packet Delivery Ratio.

The selected performance metrics are summarized in Table X.

X. INPUT ANALYSIS

Input analysis is used to explain the traffic generation models and simulation input parameters used in the experiments. In this project, the input values are not taken from a real smart home trace file. Instead, they are selected according to the meaning of each sensor type and the aim of each scenario. Because of this, the input analysis is based on mathematical and application based explanation. It is not based on formal distribution fitting.

In the Base Scenario, temperature and humidity sensors represent routine environmental monitoring devices. These sensors are normally expected to send measurements at regular times. Because of this, their packet generation is modelled with a deterministic 1 hour interval. However, if all periodic

sensors start sending packets at the same time, the simulation may create artificial synchronization. This would not represent realistic independent device behaviour. For this reason, the first packet time of temperature and humidity sensors is randomized by using `uniform(0h, 1h)`. This spreads the first transmissions over the first hour and reduces artificial simultaneous packet generation.

Smoke alarm sensors and door sensors represent event based devices. Their packet generation times are not periodic in real life. Smoke alarms and door events happen randomly over time. Because of this, exponentially distributed inter arrival times are used for these sensors in the Base Scenario. Smoke alarm sensors use `exponential(24h)`. This represents rare emergency alarm events. Door sensors use `exponential(4h)`. This represents more frequent security related events. This modelling choice is suitable for independent random arrivals over time.

In the Emergency Constant Burst Scenario, the input parameters are changed on purpose to create a high traffic condition during the full 3 day simulation period. The number of nodes and gateways stays the same. However, packet generation intervals are reduced. In this scenario, all nodes use `uniform(0s, 10s)` for the first packet time. They also use `exponential(300s)` for the next packet interval. Emergency nodes, which are Nodes 2, 3, 4 and 5, use a stronger burst configuration. Their first packet time is `uniform(0s, 2s)`. Their next packet interval is `exponential(15s)`. This creates a constant emergency burst condition. It also helps the simulation test the LoRa gateway and wireless channel under heavier traffic load.

No formal goodness of fit test, Q-Q plot or real trace validation is used in this project. This is because the simulation inputs are scenario based modelling assumptions. They are not fitted values from collected real world data. For this reason, the selected distributions are explained by the expected behaviour of the sensors and by the purpose of the scenarios.

The input variables, analysis methods and conclusions are summarized in Table XI.

XI. RESULTS AND OUTPUT ANALYSIS

This section presents the simulation results for the Base Scenario and the Emergency Constant Burst Scenario. The results are analysed with Packet Delivery Ratio, End to End Delay, Total Received Packets and Gateway Queue Length. The simulations were executed for 20 independent runs. Each run used a 3 day simulation duration and a 2 hour warm up period.

A. Base Scenario Results

The Base Scenario represents normal smart home operation. It includes periodic low priority monitoring traffic and occasional high priority emergency or security traffic. In this scenario, the network delivered all generated packets successfully. As shown in Fig. 4, the sensor nodes generated 2561 packets in total. The network server also received all 2561



Fig. 4. Packet Delivery Ratio in the Base Scenario

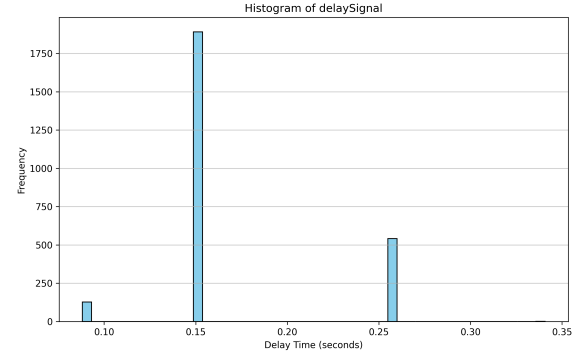


Fig. 5. Delay Histogram in the Base Scenario

packets. Because of this, the observed Packet Delivery Ratio is 100.0% over 20 simulation runs.

The delay results also show stable and low delay behaviour in the Base Scenario. The mean End to End Delay is 0.1694 seconds. The standard deviation is 0.0473 seconds. The 95% confidence interval half width is ± 0.0018 seconds. Because of this, the mean delay can be reported as 0.1694 ± 0.0018 seconds. This result shows that packets are delivered with low delay under the normal traffic configuration.

The delay histogram in Fig. 5 shows that most delay values are collected around low delay regions. The strongest group appears around 0.15 seconds. Other visible groups appear around 0.09 seconds and 0.26 seconds. This behaviour is suitable for LoRa transmission characteristics and different radio settings such as spreading factor and communication delay effects.

Fig. 6 shows the total received packet distribution across the runs. The received packet count changes between runs because some packet sources use random inter arrival times. However,

TABLE XI
INPUT ANALYSIS SUMMARY

Input Variable	Analysis Method	Conclusion
Temperature/humidity first packet time in Base Scenario	Mathematical and modelling explanation using <code>uniform(0h,1h)</code>	The uniform distribution spreads the first transmissions of routine sensors over the first hour. It also prevents artificial synchronization at the start of the simulation.
Temperature/humidity next packet interval in Base Scenario	Deterministic periodic model with 1 hour interval	Routine environmental monitoring is naturally periodic. Because of this, a fixed 1 hour interval is suitable for low priority background traffic.
Smoke alarm inter arrival time in Base Scenario	Physical explanation of rare independent emergency events using <code>exponential(24h)</code>	Smoke alarm events are random and rare. An exponential inter arrival model with a 24 hour mean represents independent emergency arrivals over time.
Door sensor inter arrival time in Base Scenario	Physical explanation of random security events using <code>exponential(4h)</code>	Door events happen randomly and more often than smoke alarms. Because of this, an exponential inter arrival model with a 4 hour mean is used.
All node first packet time in Emergency Constant Burst Scenario	Scenario based stress input using <code>uniform(0s,10s)</code>	The first packet times are distributed over a short time interval. This starts the burst condition quickly without forcing all nodes to transmit at the exact same time.
All node next packet interval in Emergency Constant Burst Scenario	Scenario based high load model using <code>exponential(300s)</code>	The shorter inter arrival time increases general traffic intensity. It creates a sustained burst condition compared with the Base Scenario.
Emergency node first packet time in Emergency Constant Burst Scenario	Scenario based emergency burst input using <code>uniform(0s,2s)</code>	Emergency nodes start generating high priority traffic almost immediately. This represents the start of an emergency heavy operating condition.
Emergency node next packet interval in Emergency Constant Burst Scenario	Scenario based high priority burst model using <code>exponential(15s)</code>	The very short inter arrival time creates sustained high priority traffic. This helps test gateway queueing and packet delivery under heavy emergency load.

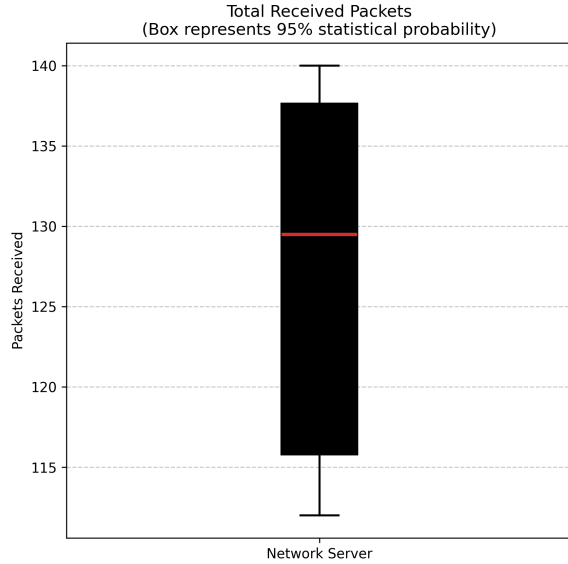


Fig. 6. Total Received Packets in the Base Scenario

the values stay in a limited range. This supports the stable traffic behaviour of the Base Scenario.

The gateway queue timeline for the Base Scenario is shown in Fig. 7. The high priority and low priority queues are mostly empty. The queue length sometimes increases to one packet. No long backlog is observed. This shows that the single shared gateway server is enough for the normal traffic load in the Base Scenario.

B. Emergency Constant Burst Scenario Results

The Emergency Constant Burst Scenario represents a heavier traffic condition. In this scenario, routine and emergency packet generation intervals are reduced. The same network topology is used, but the traffic intensity is much higher. This scenario is used to observe how the LoRa gateway and wireless channel behave under continuous emergency heavy traffic.

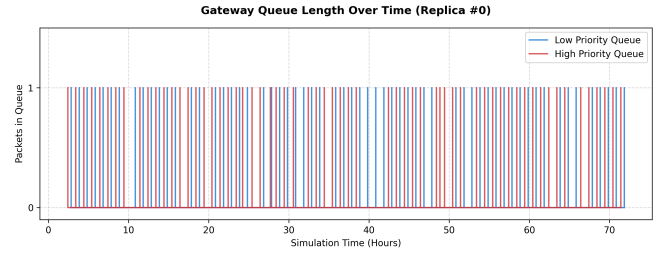


Fig. 7. Gateway Queue Length Timeline in the Base Scenario

As shown in Fig. 8, the sensor nodes generated 14194 packets in total. The network server received 8553 packets. Because of this, the Packet Delivery Ratio decreases to 60.3% over 20 simulation runs. This result shows that higher packet generation intensity causes packet loss in the emergency burst configuration.

The End to End Delay also increases in the Emergency Constant Burst Scenario. The mean delay is 2.6665 seconds. The standard deviation is 6.2952 seconds. The 95% confidence interval half width is ± 0.1334 seconds. When this result is compared with the Base Scenario, both average delay and delay variation are much higher. This is expected because the gateway and the wireless channel receive packets much more often in this scenario.

Fig. 9 shows the delay distribution for the Emergency Constant Burst Scenario. Different from the Base Scenario, delay values are spread over a much wider range. The distribution also has a long tail. Most packets still have relatively low delay. However, some packets have much larger delays because of heavier traffic and higher queueing pressure.

The received packet boxplot in Fig. 10 shows that the number of received packets changes across the runs. This variation is expected because packet generation is stochastic. Also, the emergency burst scenario uses much shorter inter arrival times. The boxplot shows why repeated runs are needed instead of using only one simulation run.

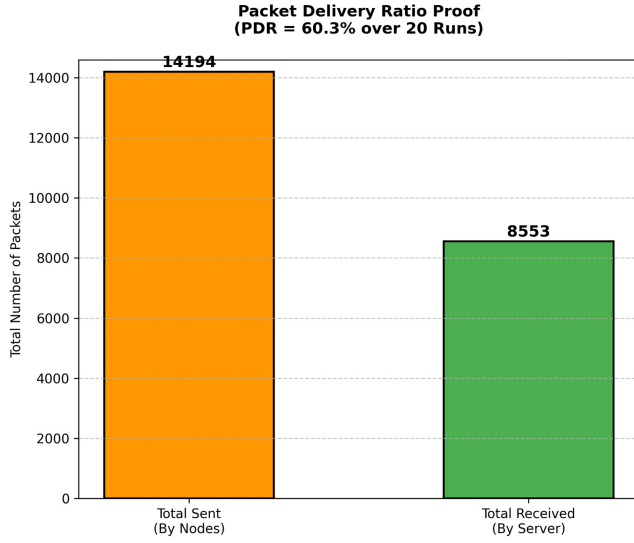


Fig. 8. Packet Delivery Ratio in the Emergency Constant Burst Scenario

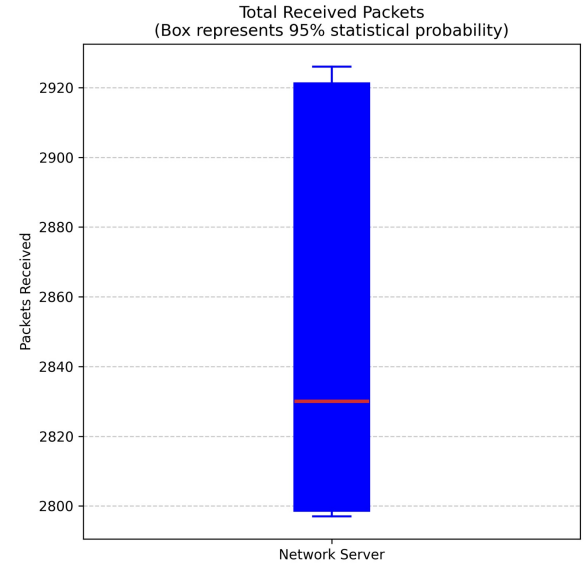


Fig. 10. Total Received Packets in the Emergency Constant Burst Scenario

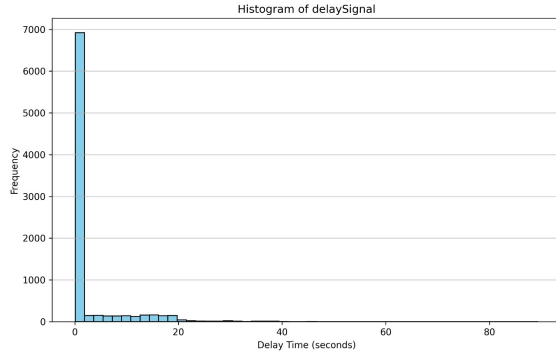


Fig. 9. Delay Histogram in the Emergency Constant Burst Scenario

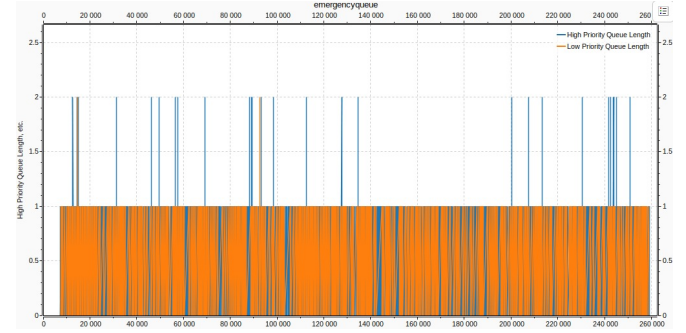


Fig. 11. Gateway Queue Length Timeline in the Emergency Constant Burst Scenario

The gateway queue timeline for the Emergency Constant Burst Scenario is shown in Fig. 11. Compared with the Base Scenario, queue activity is much denser. The low priority queue is active during a large part of the simulation. The high priority queue also sometimes reaches higher values. This shows that constant burst traffic creates stronger queueing pressure at the gateway. Since high priority packets are always served before low priority packets, the queueing mechanism gives priority to emergency traffic. However, the overall system still has packet loss and higher delay under this heavier load.

C. Comparison of Scenarios

The Base Scenario and the Emergency Constant Burst Scenario show clearly different system behaviour. In the Base Scenario, the network delivers all generated packets successfully. It also keeps a low mean delay of 0.1694 seconds. In the Emergency Constant Burst Scenario, the traffic intensity is much higher. The Packet Delivery Ratio decreases to 60.3%. The mean delay also increases to 2.6665 seconds. The standard

deviation becomes much larger too. This shows that the system is reliable under normal smart home traffic. However, its performance decreases when continuous emergency heavy traffic is applied during the full simulation period.

The main reason for this difference is the shorter packet inter arrival time in the Emergency Constant Burst Scenario. When packets are generated more often, the LoRa wireless channel and the gateway forwarding mechanism have higher load. This increases packet loss, delay and queue activity. Therefore, the results show that the gateway side priority queueing mechanism is useful for separating emergency traffic and routine traffic. However, the overall LoRa based network still has capacity limits under heavy burst conditions.

XII. DISCUSSION

The simulation results show that the LoRa based Smart Home Monitoring and Emergency Alert System works reliably in the Base Scenario. However, its performance decreases under continuous emergency heavy traffic. In the Base Scenario, the system delivered all generated packets successfully.

TABLE XII
SIMULATION RESULTS SUMMARY

Scenario	Metric	Result / Average	Std. Dev. / CI
Base Scenario	Packet Delivery Ratio	100.0% 2561 received / 2561 sent	No packet loss observed
Base Scenario	End to End Delay	Mean = 0.1694 s	Std. Dev. = 0.0473 s 95% CI = ± 0.0018 s
Base Scenario	Total Received Packets	2561 packets over 20 runs	Boxplot is used for run based variation. Exact CI is not computed
Base Scenario	Gateway Queue Length	Mostly 0 packets. Sometimes 1 packet in representative timeline	Representative time series observation. No CI is computed
Emergency Constant Burst Scenario	Packet Delivery Ratio	60.3% 8553 received / 14194 sent	Packet loss observed under heavy burst traffic
Emergency Constant Burst Scenario	End to End Delay	Mean = 2.6665 s	Std. Dev. = 6.2952 s 95% CI = ± 0.1334 s
Emergency Constant Burst Scenario	Total Received Packets	8553 packets over 20 runs	Boxplot is used for run based variation. Exact CI is not computed
Emergency Constant Burst Scenario	Gateway Queue Length	Denser queue activity than the Base Scenario	Representative time series observation. No CI is computed

It achieved 100.0% Packet Delivery Ratio. The mean End to End Delay was 0.1694 seconds. The 95% confidence interval was narrow with ± 0.0018 seconds. These results show that the selected network configuration is enough for normal smart home monitoring traffic and occasional high priority events.

The Base Scenario results are suitable with the expected system behaviour. Temperature and humidity sensors generate low priority traffic only once per hour. Smoke alarm and door sensor events happen less often according to their random traffic models. Because of this, the LoRa wireless channel and the gateway forwarding mechanism are not overloaded. The queue timeline also supports this result. Both high priority and low priority queues stay mostly empty. No long backlog is observed.

In contrast, the Emergency Constant Burst Scenario creates a much heavier traffic condition. This happens because packet inter arrival times are reduced. In this scenario, the Packet Delivery Ratio decreases to 60.3%. The mean End to End Delay increases to 2.6665 seconds. The delay standard deviation also increases to 6.2952 seconds. This shows that packet delay becomes much more variable under heavy traffic. This result is expected because frequent packet generation increases the load on the shared LoRa wireless medium and the single gateway side forwarding server.

The main trade off in the project is between traffic intensity and communication performance. When packet generation intervals are long, the system can deliver packets reliably with low delay. This is seen in the Base Scenario. However, when packet generation intervals are shortened, the same network topology has packet loss, higher delay and denser queue activity. This shows that the LoRa based smart home system has clear capacity limits under continuous high load emergency traffic.

The priority queueing mechanism helps to separate emergency traffic from routine monitoring traffic. The gateway always serves the high priority queue before the low priority queue. Because of this, emergency packets are not treated in the same way as routine packets. However, the Emergency

Constant Burst Scenario shows that priority queueing alone cannot remove the effect of too much traffic. If too many packets are generated too often, the wireless channel and the gateway still experience congestion. So, priority queueing improves traffic handling, but it does not remove the capacity limits of the network.

Another important observation is that repeated simulation runs are necessary for this project. The received packet boxplots show differences between runs. This happens because packet generation includes random inter arrival times. A single simulation run could give a wrong or limited view of the system. Using 20 independent runs makes the output analysis more reliable. It also supports the confidence interval calculation for delay.

The project also has some limitations. First, the input traffic models are based on scenario based mathematical assumptions. They are not based on real smart home trace data. Because of this, the exponential distributions are explained with the physical meaning of independent random events. They are not fitted from collected real world measurements. Second, the queue timeline figures are representative time series observations. They should not be understood as confidence intervals or statistical averages for queue length. Third, the simulation uses six sensor nodes and one gateway. So, the results should not be directly generalized to larger homes or denser IoT systems without more experiments.

Overall, the results match the expected system behaviour. Under normal traffic, the LoRa based smart home system is reliable and has low delay. Under continuous emergency burst traffic, the same system has lower delivery ratio, higher delay and stronger queue pressure. These findings show that simulation is useful for finding the operational limits of IoT emergency monitoring systems before using similar designs in real deployments.

XIII. CONCLUSION

In this project, a LoRa based Smart Home Monitoring and Emergency Alert System was implemented and evaluated by using OMNeT++, INET and FLoRa. The simulated system

TABLE XIII
DISCUSSION OF MAIN FINDINGS

Finding	Technical Explanation
The Base Scenario achieved reliable packet delivery.	The Base Scenario produced 2561 sent packets and 2561 received packets. This gives 100.0% Packet Delivery Ratio. This shows that the selected LoRa network configuration is enough under normal smart home traffic.
The Base Scenario had low delay.	The mean End to End Delay was 0.1694 seconds. The 95% confidence interval was ± 0.0018 seconds. This shows stable low delay performance with routine traffic and occasional emergency events.
The Base Scenario did not have congestion in the gateway.	The gateway queue timeline shows that high priority and low priority queues were mostly empty. Only short queue increases were observed. This means that the shared gateway server was enough for the normal traffic load.
The Emergency Constant Burst Scenario reduced reliability.	The Packet Delivery Ratio decreased to 60.3% because packet generation intervals were much shorter. The larger number of generated packets increased the load on the LoRa wireless channel and forwarding mechanism.
The Emergency Constant Burst Scenario increased delay.	The mean delay increased to 2.6665 seconds. The standard deviation increased to 6.2952 seconds. This shows that heavy traffic creates more waiting, higher variation and longer delivery times.
Priority queueing improved traffic organization but did not remove capacity limits.	The gateway always serves high priority packets before low priority packets. However, under continuous burst traffic, packet loss and delay still increased. This happens because the wireless channel and the single gateway server have limited capacity.
Repeated replications were necessary.	The random packet generation process caused differences between simulation runs. Because of this, 20 independent runs were needed for a more reliable statistical interpretation.
The main limitation is the lack of real trace based input validation.	The input models were selected with mathematical and physical explanation. No real smart home trace, Q-Q plot or goodness of fit test was used. So, the results should be read as scenario based simulation findings.

has six IoT sensor nodes, one LoRa Home Gateway and one Central Controller. It also has a shared LoRa wireless channel and a priority queueing mechanism at the gateway side. Temperature and humidity sensors generate routine monitoring traffic with low priority. Smoke alarm sensors and door sensors generate emergency and security related traffic with high priority. The gateway separates these packets into high priority and low priority queues. It always serves the high priority queue first.

The project was modelled as a discrete event simulation. Packet generation, wireless transmission, gateway queue insertion, packet forwarding and packet reception were represented as simulation events. Random traffic behaviour was added by using exponential inter arrival times for smoke alarm and door sensor packets. Periodic monitoring traffic was used for temperature and humidity sensors. The experiments were executed with 20 independent runs. Each run used a 3 day simulation duration and a 2 hour warm up period.

The results show that the system works reliably under normal smart home traffic conditions. In the Base Scenario, the system generated 2561 packets and received all 2561 packets. This gives a Packet Delivery Ratio of 100.0%. The mean End to End Delay was 0.1694 seconds. The 95% confidence interval was ± 0.0018 seconds. The gateway queue timeline also showed no long backlog. This means that the gateway was enough for the base traffic load.

In the Emergency Constant Burst Scenario, the system performance decreased because packet generation intervals were much shorter. In this scenario, the Packet Delivery Ratio decreased to 60.3%. The mean End to End Delay increased to 2.6665 seconds with a 95% confidence interval of ± 0.1334 seconds. These results show that continuous high traffic creates packet loss, higher delay and stronger queueing pressure. Because of this, the simulation shows that the LoRa based smart home system is reliable under normal conditions. However, it has capacity limits under heavy emergency burst traffic.

The main limitation of this study is that the input traffic

models are based on mathematical and scenario based assumptions. They are not based on real smart home trace data. Also, the queue length timelines are used as representative visual observations. They are not used as full statistical confidence interval analyses for queue length. The system is evaluated with six sensor nodes and one gateway. So, the results should not be directly generalized to larger systems without more experiments.

As future work, the system can be extended by increasing the number of sensor nodes. Multiple LoRa gateways can also be added. Additional traffic configurations can be compared and adaptive traffic control mechanisms can be evaluated. Energy consumption analysis can also be added. This would help to understand whether the system is suitable for low power IoT smart home deployments.

CODE AVAILABILITY

The complete working project source code, OMNeT++ simulation files, NED files, C++ source and header files, INI configuration files, result-processing scripts, generated result figures, and final report PDF are publicly available at:

<https://github.com/muratdgunes/System-Simulation-Project/>

INDIVIDUAL CONTRIBUTIONS

The project was completed collaboratively by both group members. Each member contributed to the implementation, simulation execution, result processing, and report preparation. The individual contributions are summarized in Table XIV.

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TABLE XIV
INDIVIDUAL CONTRIBUTIONS

Name	Student ID	Contribution
Murat Doruk Güneş	2453272	Contributed to network setup, sensor traffic generation logic, stochastic packet generation for smoke alarm and door sensor events, simulation scenario configuration, simulation runs, output processing, statistical analysis, and final report preparation.
Berkay Ceylan	2453041	Contributed to FLoRa integration, gateway-side priority queueing mechanism, high-priority and low-priority queue logic, metric recording support, simulation runs, output processing, statistical analysis, figure preparation, and final report preparation.

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